

Week 2, Day 1 — student handout

Umesh Indrajith · Keep this sheet. SMS still starts week 4. No slices today.

TODAY'S OUTCOMES

- var, :=, const (and a light look at iota).
- Zero values: 0, "", false, nil.
- Explicit conversions — Go will not convert numbers for you.
- fmt, strings, bytes vs runes, a little time.
- First go test on func FullName(first, last string) string.

FOLDER FOR TODAY

```
mkdir -p ~/epic-go/week02/names  
cd ~/epic-go/week02
```

VS Code: File → Open Folder → ~/epic-go/week02. gofmt on save is still required.

DECLARE

```
var campus string = "Epic Learn" // package or function  
seats := 30 // function only; type inferred  
seats = 31 // already exists – use =  
const institute = "Epic Learn" // compile-time; cannot assign later
```

ZERO VALUES AND CONVERSION

```
var n int // 0  
var s string // ""  
var ok bool // false  
year := 2026  
fmt.Println(float64(year) + 0.5) // year + 0.5 does not compile  
fmt.Println(int(3.9)) // 3 – truncation is your problem
```

FMT VERBS AND STRINGS

```
fmt.Printf("name=%s n=%d ok=%t T=%T q=%q\n", "Ada", 1, true, 1, "Ada")  
strings.TrimSpace(" Ada ")  
strings.ToUpper("sms")  
strings.Contains("Ada Lovelace", "Ada")
```

BYTES VS RUNES

```
len("café")           // 5 (bytes)
utf8.RuneCountInString("café") // 4 (letters / code points)
utf8.RuneCountInString("සිංහල") // 5
```

len counts bytes (15 for සිංහල). A rune is one Unicode code point. සි is two runes: the letter plus the vowel sign. Do not index a string with s[0] and call it a letter.

TIME (PREVIEW FOR SMS DATES)

```
time.Now().Format("2006-01-02") // layout is 2006-01-02, not YYYY-MM-DD
time.Parse("2006-01-02", "2001-05-17")
```

CLI + FIRST TEST

```
go run greet.go -first Ada -last Lovelace

cd ~/epic-go/week02/names
go mod init epiclearn/week02 # ritual today; meaning in week 4
go test
```

flag.String returns a pointer. Print *first, not first. FullName homework: TrimSpace, one space, no leftover spaces if a part is empty.

WORDS TO REMEMBER

Term	Meaning today
Zero value	What you get if you do not assign: 0, "", false, nil
:=	Declare and assign, inside a function only
Conversion	You write float64(n). The compiler will not guess
byte / rune	uint8 / int32 (a Unicode code point)
go test	Runs functions named Test... in *_test.go

IF IT BREAKS

- := at top of file → use var
- int + float64 → convert one side
- prints 0xc000... → you forgot *
- go test no module → go mod init
- YYYY-MM-DD parse fails → layout is 2006-01-02

HOMEWORK (BEFORE NEXT CLASS)

1. Ten items on the homework sheet: nine short answers in exercises.md.

2. FullName + green go test including the trim cases.